22 A cylindrical copper wire of radius r, length L and resistivity ρ is connected across a constant voltage source of e.m.f. E with negligible internal resistance. The wire contains n free electrons per unit volume, each carrying a charge e.

Which of the following expressions correctly represents the drift velocity of the electrons in the wire?

A $\dfrac{E}{ne\rho L}$ **B** $\dfrac{E(\pi r^2)}{ne\rho L^3}$ **C** $\dfrac{EL}{ne\rho(\pi r^2)^2}$ **D** $\dfrac{E}{ne\rho L(\pi r^2)}$